

Introduction & Concepts of Economic Zoology

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Introduction

- Economic zoology deals with the application of zoological knowledge for the benefit of mankind.
- It includes culturing animals for mass production for human use and to control or eradicate animals that are injurious to man directly or indirectly.
- The following chapters will be discussed here:
 - Parasitic Protozoa and human diseases
 - Beneficial and harmful insects
 - Apiculture (Bee-keeping)
 - Honey bee
 - Sericulture (Silk worm)
 - Lac culture (Lac insect)
 - Pearl culture and mollusc culture, Prawn culture
 - Venomous animals – Centipedes, wasps

Beneficial Insects

- Pollinators of crops (Bees, wasps, butterflies, moths, hoverflies, beetles).
- Many plants depend on insects to transfer pollen as they forage.
- Plants attract insects in various ways, by offering pollen or nectar meals and by guiding them to the flower using scent and visual cues.
- This has resulted in strong relationships between plants and insects.
- Value of crop production from pollination by native insects is estimated to be about \$3 billion in US alone.

Neutral Insects

- Majority of insects, almost 98% of all insect species, live in low populations in different ecosystems, without causing appreciable damage. But they form an important component of the food web and work unnoticed.
- Conservation of such fauna is important since we do not know the interactions of such insects with the animal and plant species.

Injurious Insects

- Less than 1% of insects are regarded as pests. They can be classified into pests of agriculture and forestry (Locusts, caterpillars, bugs, hoppers, aphids etc.).
- Locusts are among the most destructive of all insect pests.
- Swarms of desert locusts were among the plagues of the Biblical Egyptians, and they still plague farmers throughout Asia and Africa.
- Their threat is so great that regional and international organizations monitor desert locust populations and launch control measures when necessary.

Common Household pests

- Common household pests include ants, termites, bed bugs, carpet beetles, furniture beetles, book lice, house flies, fleas, cockroaches, silver fish, clothes moths and spiders – the list seems almost endless.
- Household pests can also be a threat to health of our families by spreading bacteria, diseases or allergens in our homes.
- They can be controlled by spraying insecticides or by fumigants and by maintaining hygiene.
- Mosquitoes can spread diseases such as malaria, yellow fever, dengue fever.

Apiculture

- “Apiculture is the scientific method of rearing honeybees. “ The word 'apiculture' comes from the Latin word 'apis' meaning bee.
- Honeybees are indigenous to the Eurasian and African continents and were introduced to the Americas and Australia by European settlers.
- In India the genus *Apis* has the following species: the western honey bee, *Apis mellifera*, the eastern honey bee, *Apis cerana indica*, the rock bee, *Apis dorsata* and the small bush bee *Apis florea*.

Sericulture

- Sericulture was first introduced into China by Hoshomin, the Queen of China.
- For a long time sericulture was considered to be a national secret by the Chinese and its industrial technique.
- Later, it was introduced into Europe and Japan by smuggling the secrets from China through travelling monks.

Lac culture

- Lac Insect belongs to Laccifer of superfamily Coccoidea of order Hemiptera.
- In all 22 species have been recorded under the genus Laccifer in Indian subcontinent.
- India is still being regarded as the principal lac producing country of the world.
- India produces about 65% of the world's total output.

Pearl Culture

- Pearl is produced by black lip pearl oyster or pearl oyster, *Pinctada margaritifera* and *P. vulgaris*, which grow to a size of 12-15 cm after 3 years of culture.
- Pearl oysters occur throughout the Indo-Pacific Region.
- They have been fished for almost two centuries for the mother-of-pearl industry and fine pearls were occasionally harvested from the sea.

Centipedes

- Centipedes (Latin: hundred, legs) belong to Phylum Arthropoda, Subphylum Myriapoda, Class Chilopoda.
- They have a pair of venomous claws which are modified maxillipedes that are used for defence as well as for hunting small insects and worms.
- Centipedes are carnivorous and use their poisonous head-claws to seize and paralyse their prey.
- They hunt, mostly at night. Many of the larger centipedes have an unpleasant bite and the poison of some giant species can be dangerous to humans, especially children.

Wasps

- Insects known as wasps include the parasitic wasps and the stinging wasps of family Vespidae.
- About 75,000 species of wasps are known, most of them are parasitic or predators.
- Almost every pest insect species has at least one wasp species that is a predator or parasite upon it and hence wasps are increasingly used in biological pest control.
- Wasps are characterized by two pairs of membranous wings, three pairs of legs and an ovipositor (tube for laying eggs) that may be modified as sting in sterile females.

Precautions from wasps

- If surrounded by a swarm of bees or wasps, move out of the way slowly.
- Insect repellents are effective.
- Never aim a blow at a wasps' or bees' nest or attempt to throw them because the insects will immediately attack.
- Stay away from things that attract insects, such as flowers, trees, bushes and piles of wood.
- Long sleeves, long trousers, socks, shoes and gloves help to protect you from stings.
- Close the windows in the house and the car to keep the insects out.
- Look out for insects nests in your home or garden and have them removed immediately.
- Protective gear such as mesh covers for the face can be very effective against the nuisance of the highland midge in summer for example.

Human Diseases

- Diseases that affect humans need to be diagnosed accurately before they can be treated.
- In this section, diagnosis, pathogenesis, identification of causative organism, prevention including vaccination wherever available, treatment and drugs commonly prescribed are given.
- Dengue, Japanese Encephalitis, Rabies, Epidemic Typhus, Tuberculosis, Amoebiasis, Yellow fever, Kala Azar, Filariasis, Small pox, Plague, Malaria, Cholera, AIDS.